



ApexPlanet Data Analytics Internship — Final Project

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End-to-End Data Analytics on Supermarket Sales

Transform raw transactions into actionable business insights



Scope

Data cleaning → EDA → Modeling → Dashboarding → Reporting



Duration

Internship capstone project

Problem Statement & Objectives



Data cleaning



Exploratory analysis



Extract insights using SQL



Build interactive
dashboards



Statistical analysis



Forecast sales



Segment customers



Predictive models



Executive reports

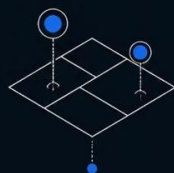
Dataset Overview



Records
(1000)



Features
(17)



Geographic
(Branches, Cities)



Product & Transaction details
(Product Lines, Customer Types,
Payment Methods, Sales)



1000 Records

Sufficient for
prototyping &
insights



17 Features

Demographics,
transaction & product
fields



Branches & Cities

Multi-branch analysis



Product Lines & Payment

7 product lines,
multiple payment
types

Data Cleaning & Exploratory Analysis

Before

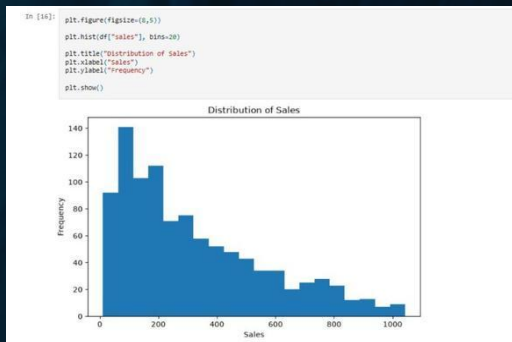
Missing values • Duplicates •
Type mismatches

After

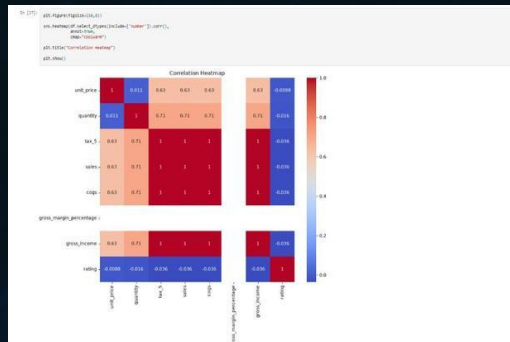
Cleaned rows • Consistent types
• Summary stats

Analysis

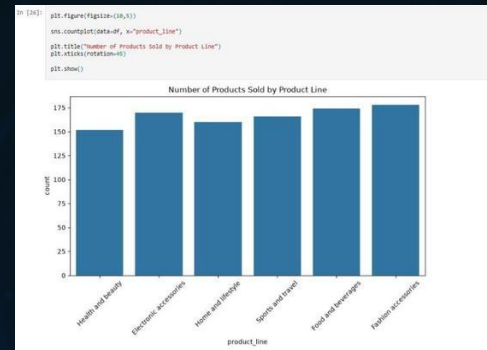
Correlation, product-line trends,
sales distribution



Sales Distribution



Correlation Heatmap



Product Line Analysis

Dashboard Component: KPI Cards

Dashboard components

322,966.75

Total Revenue

1K

Total Orders

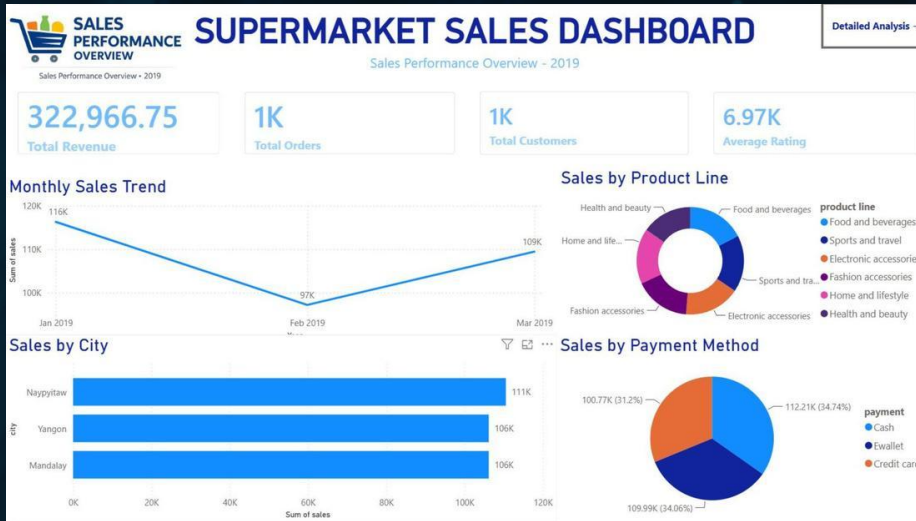
1K

Total Customers

6.97K

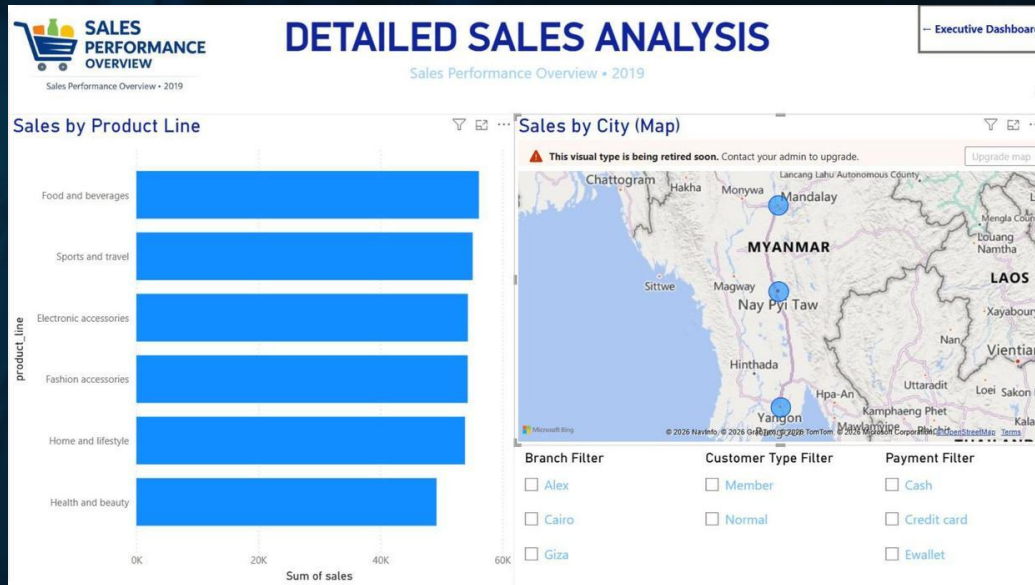
Average Rating

Dashboard Component: Product Line Analysis



A bar chart showing sales performance across different product lines with color-coded segments and trend indicators. This chart clearly visualizes the sales performance of each product line, allowing for quick identification of top performers and areas needing attention within the supermarket.

Dashboard Component: Sales by Branch



Geographic analysis showing sales distribution across different branches with interactive map filters and regional performance metrics.

Filters & Slicers

Branch Filter

☐ Alex

☐ Cairo

☐ Giza

Customer Type Filter

☐ Member

☐ Normal

Payment Filter

☐ Cash

☐ Credit card

☐ Ewallet

Interactive filter controls for date ranges, product categories, branches, and customer segments to enable dynamic dashboard exploration.

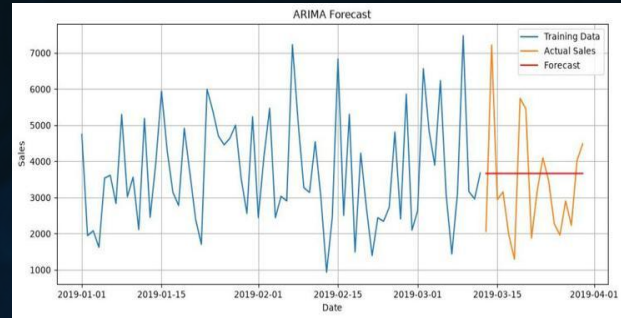
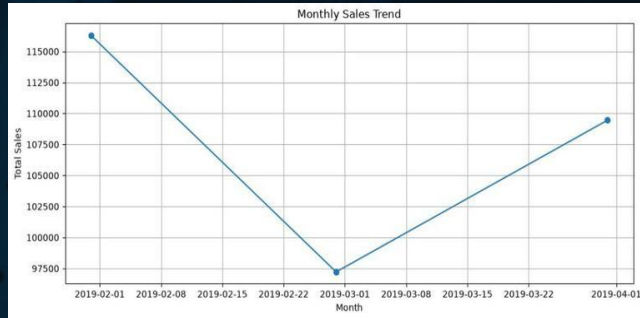
Advanced Analytics & Machine Learning

Statistical Analysis

Time Series Forecasting

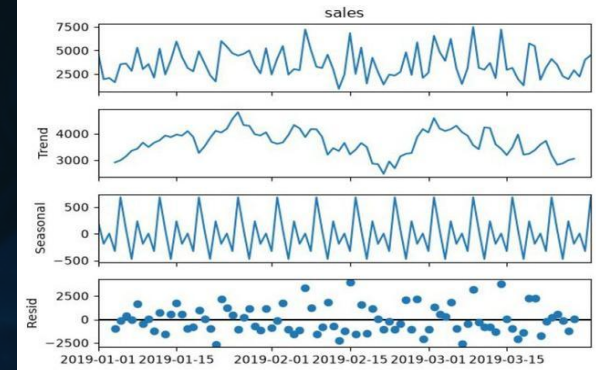
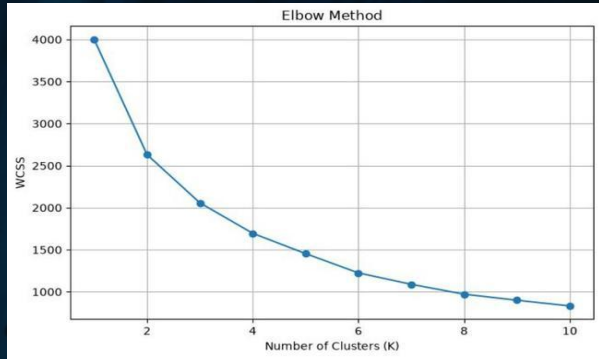
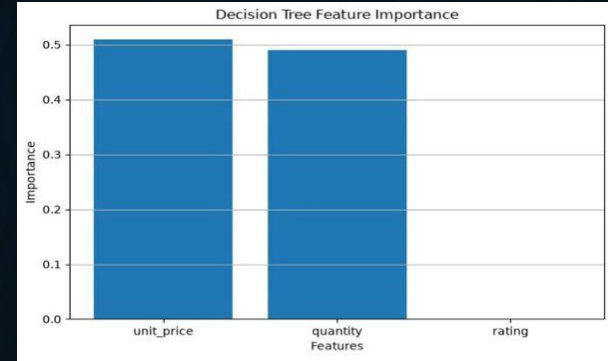
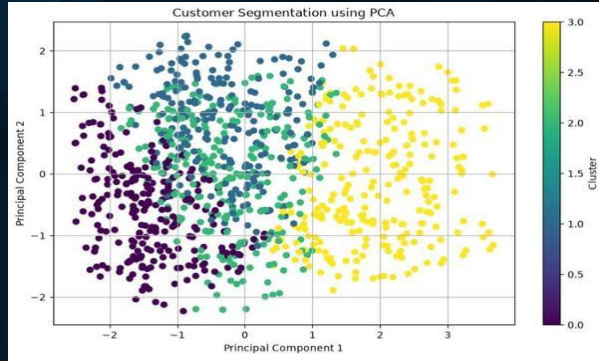
Customer Segmentation

Model Training



Monthly Sales Trend & Arima Forecast

Customer Segmentation & Decision Tree Feature Importance



Elbow Method & Sal

Business Recommendations



Improve inventory planning



Focus on high-performing product lines



Enhance customer engagement



Optimize branch performance



Use predictive analytics for planning



Automate reporting workflows

Challenges & Learnings

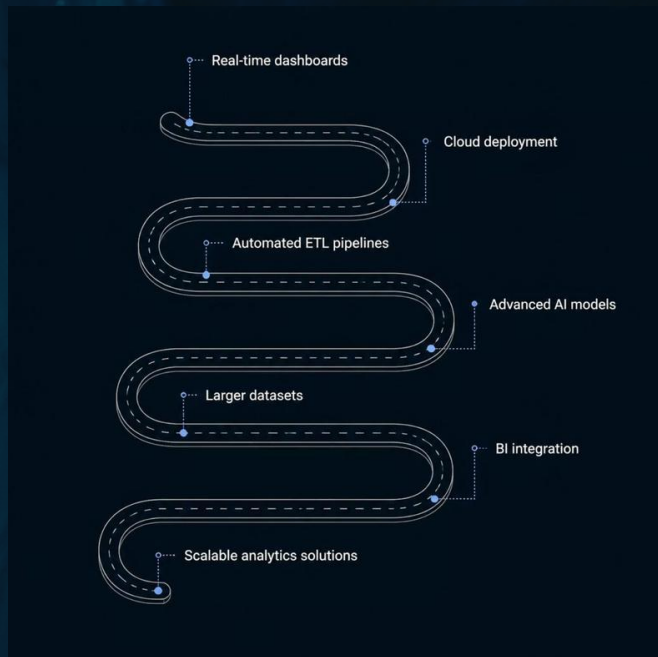
Challenges

- Data preprocessing complexity
- SQL integration with analytics pipeline
- Dashboard usability & layout
- Model evaluation & validation
- Comprehensive documentation

Learnings

- Python (pandas, scikit-learn)
- SQL (joins, window functions)
- Power BI dashboarding
- Statistics & Time-series methods
- Machine Learning & BI automation
- Version control with GitHub

Future Scope & Roadmap



Short-term

Automated ETL ·
Real-time KPIs

1

2

Mid-term

Cloud deployment ·
Scalable storage

3

Long-term

Advanced AI models ·
Enterprise BI
integration



Roadmap supports scale-up and productionization

Thank you — Questions?